

Wassim Chouayakh

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EDUCATION

National Institute of Applied Science and Technology (INSAT) – Tunis, Tunisia	Sep 2025 – Present
Engineering Degree – Industrial IT and Automation	Expected Graduation: 2028
National Institute of Applied Science and Technology (INSAT) – Tunis, Tunisia	Sep 2023 – Jun 2025
Preparatory Cycle – Mathematics, Physics and Industrial IT	

TECHNICAL SKILLS

Languages	C/C++ (bare-metal), Python, Java, Dart
Microcontrollers	STM32 F1/F4 – HAL and register-level development, interrupts, DMA, real-time
Protocols	UART, I2C, SPI, CAN Bus – register-level driver implementation
Peripherals	GPIO, Timers, PWM (generation & input capture), ADC, EXTI
Embedded Tools	STM32CubeIDE, STM32CubeMX, Keil uVision, ST-Link, Logic Analyzer
Frameworks	FastAPI, React, Next.js, Flutter
AI / ML	PyTorch, TensorFlow, scikit-learn, MLflow, CNNs, NLP, RAG
DevOps	Git, Docker, Linux/Unix, PostgreSQL, MySQL

PROJECTS

STM32 Peripheral & Protocol Lab Series STM32 F4, C, HAL, Bare-Metal, UART, I2C, CAN, ADC, PWM, DMA
- Developed 16+ bare-metal drivers (GPIO, Timers, PWM, ADC, UART, I2C) via direct register manipulation, bypassing any hardware abstraction layer - Implemented PWM input capture (frequency and duty cycle measurement) and EXTI interrupts for real-time event-driven control - Designed a full-duplex bare-metal UART driver (PC ↔ STM32) and a register-level I2C driver for external sensor communication
SightLine – AI Safety & Compliance System FastAPI, YOLOv8, HuggingFace, React, Docker, Python
- Detected PPE violations in real time with ~92% accuracy using YOLOv8 fine-tuned on 1,000+ images - Integrated a RAG engine (HuggingFace) processing 50+ regulatory PDFs; deployed full-stack platform via Docker, reducing deployment time by ~60%
CNN – Antenna Array Configuration Prediction PyTorch, MobileNetV2, Fine-tuning, NumPy, SciPy
- Fine-tuned a MobileNetV2 CNN on 2,000+ synthetic samples to predict 35-class multi-label antenna configurations, with accuracy confirmed via HPBW and side-lobe level metrics - Architected a complete end-to-end pipeline covering electromagnetic data generation, model fitting, inference and radiation pattern reconstruction, benchmarked across 35 output classes

WORK EXPERIENCE

Software Engineering Intern – KPIT, Sfax, Tunisia	Jul – Aug 2025
- Engineered a Python test-case compliance script, cutting manual review time by ~40%; automated reports reduced processing from hours to minutes - Delivered a cross-platform desktop application (Flutter + Python backend) adopted by 10+ internal testers	
AI Engineering Intern – Infosquare, Sfax, Tunisia	Jun – Jul 2025
Deployed a RAG-based chatbot (LLM + retrieval over 100+ files) improving response relevance by ~30%	

LANGUAGES

French: Native | **Arabic:** Native | **English:** Fluent (professional) | **Spanish:** Beginner